

THE SHORTEST HARMONIC SUMS WITH DECREASING DENOMINATOR

WOUTER VAN DOORN

ABSTRACT. For a positive integer a , let $b(a)$ be the smallest integer $b > a$ such that the denominator of $\frac{1}{a} + \frac{1}{a+1} + \cdots + \frac{1}{b}$ is smaller than the denominator of $\frac{1}{a} + \frac{1}{a+1} + \cdots + \frac{1}{b-1}$. Recently it was shown that the limit inferior

$$\liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right)$$

exists and is positive. Here we find its exact value.

1. INTRODUCTION

For positive integers $a < b$, let $u_{a,b}$ and $v_{a,b}$ be the coprime positive integers for which

$$\sum_{i=a}^b \frac{1}{i} = \frac{u_{a,b}}{v_{a,b}}.$$

For a given $a \in \mathbb{N}$, let $b(a)$ be the smallest integer $b > a$ for which $v_{a,b} < v_{a,b-1}$. Whether such a $b(a)$ exists for all a was asked by Erdős and Graham in [4], recorded as Erdős Problem #290 at Bloom's website [1], and resolved by the author in [3], with the upper bound $b(a) \leq 4.374(a-1)$ for all $a \geq 6$.

As for lower bounds on $b(a)$, in [3] it was also proved that $b(a) - a \gg \log a$ holds for all a , but that there are infinitely many a with $b(a) - a \ll \log a$. To be a bit more precise, for a positive integer d , let us define the polynomial

$$f_d(x) := \sum_{i=0}^d \prod_{\substack{j=0 \\ j \neq i}}^d (x - j),$$

and let $\delta(f_d)$ be the density of primes p such that $f_d(x) \equiv 0 \pmod{p}$ is solvable. This density exists by Chebotarev's density theorem, and the values of $\delta(f_d)$ depend on specifics of the Galois group of $f_d(x)$. For more information on this we refer to [3, Section 3.3]. With

$$c := \sum_{d=1}^{\infty} \frac{\delta(f_d)}{d(d+1)},$$

there it was also proved that we have the string of inequalities

$$0.54 < \frac{1}{1+c} \leq \liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right) \leq \frac{1}{2c} < 0.61.$$

This paper is dedicated to showing that the lower bound is sharp.

Theorem 1. *We have*

$$\liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right) = \frac{1}{1 + c}.$$

We will actually prove the slightly stronger statement that for all $C < 1 + c$ there exists an N such that for all $n \geq N$ there exist integers $a, b > e^{Cn}$ with $b = a + n$ for which $v_{a,b} < v_{a,b-1}$. We further remark that $\frac{1}{1+c}$ is approximately equal to 0.546, although we make no effort here to calculate it as precisely as possible. Even so, if we combine Theorem 1 with [3, Lemma 32], then we find that there are infinitely many a and b with

$$a < b < a + 0.55 \log a \quad \text{and} \quad v_{a,b} < v_{a,b-1}.$$

2. REDUCTION OF THE MAIN RESULT

In this section we reduce the proof of Theorem 1 to the existence of a certain integer x . This reduction already follows from the arguments in [3, Section 3.3], but in order to make this paper self-contained, we give the proof here in full. To set this all up, we need a couple of definitions.

Let $D \geq 2$ be an integer and let n be an integer that is sufficiently large in terms of D . For a positive integer $d \leq 2D$, let S_d be the set of primes p with $\frac{n}{d+1} < p \leq \frac{n}{d}$ such that $f_d(x) \equiv 0 \pmod{p}$ is solvable.

Example 2. *By expanding out the definition of $f_d(x)$ for $d = 2$, we find $f_2(x) = 3x^2 - 6x + 2$. For this polynomial, one can check that an odd prime $q \in (\frac{n}{3}, \frac{n}{2}]$ belongs to S_2 precisely when $q \equiv \pm 1 \pmod{12}$, in which case $f_2(x)$ actually has two zeros modulo q . Moreover, since $\frac{n}{d} - \frac{n}{d+1} = \frac{n}{d(d+1)}$, by the prime number theorem in arithmetic progressions we find*

$$|S_2| = \left(\frac{1}{12} + o(1) \right) \frac{n}{\log n},$$

when n goes to infinity.

We now define Q as the product of all primes $q \in \bigcup_{d \leq 2D} S_d$ and, for a prime factor q of Q , set $Q_q := \frac{Q}{q}$.

Similarly, for a positive integer $d \leq D$, we let T_d be the set of primes p with $\frac{n}{d+1} < p \leq \frac{n}{d}$ such that $f_d(x) \equiv 0 \pmod{p}$ is *not* solvable, define P as the product of all primes $p \in \bigcup_{d \leq D} T_d$ and set $P_p := \frac{P}{p}$ for a prime $p \mid P$. We then claim that it is sufficient to find an x with $\frac{Q}{n} \leq x < Q$ satisfying various congruences.

Theorem 3. *Assume that for all integers $D \geq 2$ and all sufficiently large integers n there exists an integer $x < Q$ with the following three properties:*

- (1) *We have $x > \frac{Q}{n}$.*
- (2) *For all $d \leq 2D$ and all $q \in S_d$ we have $f_d(xPQ_q) \equiv 0 \pmod{q}$.*
- (3) *For all $d \leq D$ and all $p \in T_d$ we have $f_{d-1}(xP_pQ - 1) \not\equiv 0 \pmod{p}$.*

Then

$$\liminf_{a \rightarrow \infty} \left(\frac{b(a) - a}{\log a} \right) \leq \frac{1}{1 + c}.$$

Proof. Let $D \geq 2$ be a fixed integer, let n be sufficiently large in terms of D , and let x be as in the theorem statement. We then define $b := xPQ$ and $a := b - n$, and need to show that we have

$$b \leq a + \left(\frac{1}{1+c} + o(1) \right) \log a \quad \text{and} \quad v_{a,b} < v_{a,b-1}.$$

As for the first inequality, we apply the fact that the densities $\delta(f_d)$ exist for fixed d , which by the prime number theorem give

$$\begin{aligned} P &= \exp \left[\left(\sum_{d=1}^D \frac{1 - \delta(f_d)}{d(d+1)} + o(1) \right) n \right], \\ Q &= \exp \left[\left(\sum_{d=1}^{2D} \frac{\delta(f_d)}{d(d+1)} + o(1) \right) n \right], \\ x &= Q^{1+o(1)} = \exp \left[\left(\sum_{d=1}^{2D} \frac{\delta(f_d)}{d(d+1)} + o(1) \right) n \right]. \end{aligned}$$

Here, the $o(1)$ terms go to 0 as n goes to infinity. By then letting D go to infinity as well, we find

$$b = xPQ = \exp \left[(1 + c + o(1))n \right],$$

from which the desired inequality follows.

It remains to show that the above definitions of a and b imply $v_{a,b} < v_{a,b-1}$. For this, we make use of the p -adic valuation ν_p . Since $x < Q$ by assumption, in order to show $v_{a,b} < v_{a,b-1}$ it suffices to show that the two inequalities

$$\nu_q(v_{a,b}) < \nu_q(v_{a,b-1}) \quad \text{and} \quad \nu_p(v_{a,b}) \leq \nu_p(v_{a,b-1}) + \nu_p(x)$$

hold for all $q \mid Q$ and $p \nmid Q$ respectively. Indeed, when multiplying over all primes, these two sets of inequalities imply $\frac{v_{a,b}}{v_{a,b-1}} \leq \frac{x}{Q} < 1$, finishing the proof. Hence, let us first show that $\nu_q(v_{a,b}) < \nu_q(v_{a,b-1})$ holds for all $q \mid Q$.

Let q be an arbitrary prime factor of Q , assume without loss of generality $q \in S_d$, and recall the definition of $f_d(x)$. For an integer $i \in \{0, 1, \dots, d\}$ we see that all terms in $f_d(i)$ vanish, except for the i th term. In particular, as

$$q > \frac{n}{2D+1} > 2D \geq d$$

if n is sufficiently large, we find

$$f_d(i) = \prod_{\substack{j=0 \\ j \neq i}}^d (i-j) \not\equiv 0 \pmod{q}.$$

We therefore conclude from the second property that $xPQ_q \not\equiv i \pmod{q}$ for all $i \in \{0, 1, \dots, d\}$. This implies that, among the multiples $b, b-q, \dots, b-dq$ of q in the interval $[a, b]$, none of these multiples are divisible by q^2 , since

$$\frac{b-iq}{q} \equiv xPQ_q - i \not\equiv 0 \pmod{q}.$$

By multiplying the partial harmonic sum by q , all terms vanish modulo q except for the terms with denominator equal to one of $b, b - q, \dots, b - dq$. Hence,

$$q \sum_{i=a}^b \frac{1}{i} \equiv \sum_{i=0}^d \frac{1}{xPQ_q - i} \equiv \frac{f_d(xPQ_q)}{\prod_{i=0}^d (xPQ_q - i)} \equiv 0 \pmod{q},$$

so that $\nu_q(v_{a,b}) = 0$. On the other hand, omitting the final term $\frac{1}{b}$ in the previous calculation gives

$$q \sum_{i=a}^{b-1} \frac{1}{i} \equiv -\frac{1}{xPQ_q} \not\equiv 0 \pmod{q},$$

which implies $\nu_q(v_{a,b-1}) \geq 1$, as desired. In fact, $\nu_q(v_{a,b-1})$ is exactly equal to 1, since we just saw that none of the multiples of q in $[a, b]$ are divisible by q^2 .

To finish the proof, we need to show that

$$\nu_p(v_{a,b}) \leq \nu_p(v_{a,b-1}) + \nu_p(x)$$

holds for every prime $p \nmid Q$. This is not hard to see for $p \nmid P$, as we then get $\nu_p(b) = \nu_p(x)$, while the inequalities

$$\nu_p(v_{a,b}) \leq \max(\nu_p(v_{a,b-1}), \nu_p(b)) \leq \nu_p(v_{a,b-1}) + \nu_p(b)$$

hold in general. Hence, for the rest of the proof we may assume $p \mid P$, so let $p \in T_d$, say.

As before, the multiples of p in the interval $[a, b]$ are exactly $b, b - p, \dots, b - dp$. If $xP_pQ \not\equiv i \pmod{p}$ for all $0 \leq i \leq d$, then none of those multiples of p are divisible by p^2 , so that the third property gives

$$p \sum_{i=a}^{b-1} \frac{1}{i} \equiv \sum_{i=1}^d \frac{1}{xP_pQ - i} \equiv \frac{f_{d-1}(xP_pQ - 1)}{\prod_{i=1}^d (xP_pQ - i)} \not\equiv 0 \pmod{p}.$$

In particular, $\nu_p(v_{a,b-1}) = 1 \geq \nu_p(v_{a,b})$, which suffices. On the other hand, if xP_pQ is congruent to $i \pmod{p}$ for some $0 \leq i \leq d$, then $b - ip$ is divisible by p^2 . Moreover, since $p > \frac{n}{d+1}$ implies $p > d$ as long as n is large enough, $b - ip$ is then the unique integer in the interval $[a, b]$ divisible by p^2 . In this case, $i \neq 0$ implies

$$\nu_p(v_{a,b}) = \nu_p(b - ip) = \nu_p(v_{a,b-1}) \leq \nu_p(v_{a,b-1}) + \nu_p(x),$$

while $i = 0$ implies $\nu_p(v_{a,b-1}) = 1$ by the third property again, so that

$$\nu_p(v_{a,b}) = \nu_p(b) = \nu_p(P) + \nu_p(x) = 1 + \nu_p(x) = \nu_p(v_{a,b-1}) + \nu_p(x). \quad \square$$

3. PROOF OF THE MAIN RESULT

In this section we prove Theorem 1 by constructing an x for which the three properties in Theorem 3 hold. As the lower bound on the liminf of $\frac{b(a)-a}{\log a}$ is precisely [3, Lemma 31], this is sufficient.

Proof of Theorem 1. Let $D \geq 2$ be given, let n be sufficiently large in terms of D , and let the products P and Q be as before. We define the following zeros of f_d (or a transformed version thereof) for prime factors of P and Q :

- For every $q \in S_2$, let $x_q^{(1)}$ and $x_q^{(2)}$ denote the two distinct zeros of $f_2(x) \pmod{q}$.

- Let $x_{q_0}^{(1)}, \dots, x_{q_0}^{(2D)}$ denote the $2D$ distinct zeros of $f_{2D}(x) \pmod{q_0}$ for some specific $q_0 \in \left(\frac{n}{2D+1}, \frac{n}{2D}\right]$ for which f_{2D} splits completely modulo q_0 . The existence of such a q_0 is guaranteed by Chebotarev's density theorem, assuming n is large enough.
- For every $d \leq 2D$ different from 2 and every $q \in S_d$ different from q_0 , let $x_q^{(1)}$ be any zero of $f_d(x) \pmod{q}$.
- For every $d \leq D$ and every $p \in T_d$, let $x_p^{(1)}, \dots, x_p^{(e_p)}$ denote the distinct values in $[1, p]$ for which $f_{d-1}(x_p^{(i)} P_p Q - 1) \equiv 0 \pmod{p}$ for all i .

Remark 4. For all $d \leq D$ and all $p \in T_d$ we have $e_p < D$. Indeed, f_{d-1} has degree $d-1$, so if n is large enough, then $f_{d-1} \pmod{p}$ has at most $d-1 < D$ roots for any $p \in T_d$.

For $1 \leq k \leq 2D$, let $x_0^{(k)}$ be the smallest positive integer with, for all $q \mid Q$ different from q_0 ,

$$x_0^{(k)} \equiv x_q^{(1)} (PQ_q)^{-1} \pmod{q} \quad \text{and} \quad x_0^{(k)} \equiv x_{q_0}^{(k)} (PQ_{q_0})^{-1} \pmod{q_0}.$$

These $x_0^{(1)}, \dots, x_0^{(2D)}$ are our initial potential values for x and they ensure the second property. Essentially all we will do from here is change the congruence

$$x \equiv x_q^{(i)} (PQ_q)^{-1} \pmod{q} \quad \text{to} \quad x \equiv x_q^{(i')} (PQ_q)^{-1} \pmod{q}$$

for various $q \in S_2 \cup \{q_0\}$ to obtain a final x for which the first and third properties hold as well. Now, for a $q \in S_2$, let $t_q \in [1, q)$ be the smallest positive integer with

$$t_q Q_q \equiv \left(x_q^{(2)} - x_q^{(1)}\right) (2PQ_q)^{-1} \pmod{q}.$$

Then switching from $x_q^{(1)}$ to $x_q^{(2)}$ changes x by $2t_q Q_q \pmod{Q}$. These changes for $q \in S_2$ in principle give $2^{|S_2|}$ values of x to work with, but we will not need to consider all of them.

Let S'_2 be equal to S_2 but with the largest prime removed if $|S_2|$ is odd, and further define $x_1^{(k)}$ for a positive integer $k \leq 2D$ to be the unique integer in $[1, Q]$ with

$$x_1^{(k)} \equiv x_0^{(k)} + \sum_{q \in S'_2} t_q Q_q \pmod{Q}.$$

In a sense, $x_1^{(k)}$ is the halfway point between making all swaps for $q \in S'_2$. We now order the primes $q \in S'_2$ according to the value of $t_q Q_q$. That is, we write $S'_2 = \{q_1, q_2, \dots, q_{|S'_2|}\}$ with $t_{q_i} Q_{q_i} < t_{q_j} Q_{q_j}$ if $i < j$. We note that these values of $t_{q_i} Q_{q_i}$ are all distinct, as otherwise such a value would be divisible by both Q_{q_i} and Q_{q_j} for distinct i, j . However, this would imply that $t_{q_i} Q_{q_i}$ is divisible by Q , which is impossible as $1 \leq t_{q_i} Q_{q_i} < Q$, by definition of t_{q_i} .

For every odd $i < |S'_2|$ we only consider those values of x we obtain by switching exactly one of q_i and q_{i+1} to its second root. With

$$\Delta_i := t_{q_{i+1}} Q_{q_{i+1}} - t_{q_i} Q_{q_i} > 0,$$

switching q_i to its second root decreases the residue by Δ_i relative to $x_1^{(k)}$, while switching q_{i+1} to its second root increases it by Δ_i . If we consider all these paired

changes, then the final result is congruent to

$$x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \pmod{Q},$$

where the $\varepsilon_i \in \{-1, 1\}$ can be chosen arbitrarily. Moreover,

$$\left| \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \right| \leq \sum_{i < |S'_2| \text{ odd}} \Delta_i \leq \sum_{i=1}^{|S'_2|-1} (t_{q_{i+1}} Q_{q_{i+1}} - t_{q_i} Q_{q_i}) < Q,$$

due to the fact that the final sum telescopes and $1 \leq t_q Q_q < Q$ for all q . We now need two lemmas on the prime factors of Δ_i and $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$.

Lemma 5. *For all but at most two primes $p \mid P$, at least $\frac{1}{6}|S'_2|$ of the Δ_i are non-zero modulo p .*

Proof. Since

$$\Delta_i = t_{q_{i+1}} Q_{q_{i+1}} - t_{q_i} Q_{q_i} = \frac{Q}{q_i q_{i+1}} (q_i t_{q_{i+1}} - q_{i+1} t_{q_i}),$$

a prime $p \mid P$ divides Δ_i precisely when p divides $q_i t_{q_{i+1}} - q_{i+1} t_{q_i}$. This latter integer is smaller than n^2 , while every prime divisor of P is larger than $\frac{n}{D+1}$. Hence, if n is large enough, there can be at most two prime factors of P that divide $q_i t_{q_{i+1}} - q_{i+1} t_{q_i}$. As there are a total of $\frac{1}{2}|S'_2|$ values of Δ_i to consider, we deduce

$$\sum_{p \mid P} \sum_{\substack{i < |S'_2| \text{ odd} \\ p \mid \Delta_i}} 1 \leq |S'_2|.$$

It follows that there are at most two exceptional primes $p \mid P$ for which more than $\frac{1}{3}|S'_2|$ of the Δ_i vanish modulo p . For every other $p \mid P$, at least $\frac{1}{6}|S'_2|$ of the Δ_i are non-zero modulo p . $\square$

Let p be a non-exceptional prime divisor of P . We then define R_p to be the number of quadruples $(i, j, \varepsilon_i, \varepsilon_j)$ for which p divides $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$, where $i, j \in \{1, 3, \dots, |S'_2| - 1\}$ and $\varepsilon_i, \varepsilon_j \in \{-1, 1\}$.

Lemma 6. *We have*

$$\sum_{\substack{p \mid P \\ p \text{ non-exceptional}}} R_p \leq 19|S'_2|^2.$$

Proof. We note that $\sum_p R_p$ is equal to the number of quintuples $(i, j, \varepsilon_i, \varepsilon_j, p)$ for which p divides $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$. Now, if a prime p divides $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$, then p either divides both Δ_i and Δ_j , or neither. By using Lemma 5 and its proof, the number of triples (i, j, p) where a non-exceptional prime divisor of P divides both Δ_i and Δ_j is at most

$$\sum_{\substack{p \mid P \\ p \text{ non-exceptional}}} \left(\sum_{\substack{i < |S'_2| \text{ odd} \\ p \mid \Delta_i}} 1 \right)^2 \leq \frac{1}{3}|S'_2| \sum_{p \mid P} \sum_{\substack{i < |S'_2| \text{ odd} \\ p \mid \Delta_i}} 1 \leq \frac{1}{3}|S'_2|^2.$$

Multiplying by 4 to account for the signs of ε_i and ε_j gives an upper bound for the corresponding number of quintuples $(i, j, \varepsilon_i, \varepsilon_j, p)$ where p divides both Δ_i and Δ_j .

As for the case where p divides neither Δ_i nor Δ_j , let us first consider the tuples with $i = j$. In this instance, p only divides $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$ if $\varepsilon_i = -\varepsilon_j$, so for every $p \mid P$ there are at most $|S'_2|$ such tuples. As the number of prime factors of P is at most the number of primes smaller than or equal to n , which in turn is smaller than $13|S'_2|$ by the bound in Example 2, the number of quintuples we are interested in with $i = j$ is at most $13|S'_2|^2$.

Finally, if $i \neq j$ and p divides $\varepsilon_i \Delta_i + \varepsilon_j \Delta_j$, then p must divide

$$\frac{q_i q_{i+1} q_j q_{j+1} (\varepsilon_i \Delta_i + \varepsilon_j \Delta_j)}{Q} = \varepsilon_i q_j q_{j+1} (q_i t_{q_{i+1}} - q_{i+1} t_{q_i}) + \varepsilon_j q_i q_{i+1} (q_j t_{q_{j+1}} - q_{j+1} t_{q_j}).$$

This quantity is non-zero as it is congruent to $-\varepsilon_i q_j q_{j+1} q_{i+1} t_{q_i} \not\equiv 0 \pmod{q_i}$, while its absolute value is smaller than n^4 . Hence, it has at most four prime factors $p \mid P$, once again due to $p > \frac{n}{D+1}$. As the number of quadruples $(i, j, \varepsilon_i, \varepsilon_j)$ is at most $|S'_2|^2$, we conclude that, summed over all non-exceptional prime factors of P ,

$$\sum_p R_p \leq \left(\frac{4}{3} + 13 + 4 \right) |S'_2|^2 < 19|S'_2|^2. \quad \square$$

We now choose the ε_i independently and uniformly from $\{-1, 1\}$ and apply [5, Theorem 1.4], which is an inequality of Halász [6] over $\mathbb{F}_p$. This inequality bounds the probability that a signed sum is congruent to a specific residue class in terms of the quantity R_p we just considered. More precisely, in the notation of [5], we let

$$\begin{aligned} \mathbf{a} &:= (\Delta_1, \Delta_3, \dots, \Delta_{|S'_2|-1}), \\ k &:= 1, \\ M &:= \frac{1}{180} |S'_2|, \\ n &:= \frac{1}{2} |S'_2|, \\ R_k(\mathbf{a}) &:= R_p. \end{aligned}$$

The hypotheses of [5, Theorem 1.4] are then satisfied, and we find that the probability that the sum

$$\sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i$$

is congruent to a given residue class mod p is bounded by

$$\frac{1}{p} + \frac{C R_p}{|S'_2|^{5/2}} + e^{-\frac{1}{180} |S'_2|},$$

for some absolute constant C . In particular, as $e_p < D$ for any prime factor p of P by Remark 4, we have

$$\Pr \left(\exists j \leq e_p : x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \equiv x_p^{(j)} \pmod{p} \right) < D \left(\frac{1}{p} + \frac{C R_p}{|S'_2|^{5/2}} + e^{-\frac{1}{180} |S'_2|} \right).$$

Summing over all non-exceptional $p \mid P$ and using Lemma 6 shows that the probability that there exists a non-exceptional prime $p \mid P$ and a $1 \leq j \leq e_p$ such that

$$x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \equiv x_p^{(j)} \pmod{p}$$

is smaller than

$$\begin{aligned}
& D \sum_{\substack{p|P \\ p \text{ non-exceptional}}} \left(\frac{1}{p} + \frac{CR_p}{|S'_2|^{5/2}} + e^{-\frac{1}{180}|S'_2|} \right) \\
& \leq D \sum_{\frac{n}{D+1} < p \leq n} \left(\frac{1}{p} + e^{-\frac{1}{180}|S'_2|} \right) + \frac{CD}{|S'_2|^{5/2}} \sum_{\substack{p|P \\ p \text{ non-exceptional}}} R_p \\
& \ll \frac{D \log D}{\log n} + \frac{D}{\sqrt{|S'_2|}} \\
& < \frac{1}{6D},
\end{aligned}$$

for all sufficiently large n . Now, the term $x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i$ is contained in $(-Q, 2Q)$, so in order to get a potential value of $x \in [1, Q)$, we may need to add or subtract Q . Fortunately, the above bounding argument works just as well for

$$\Pr \left(\exists j \leq e_p : \pm Q + x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \equiv x_p^{(j)} \pmod{p} \right).$$

By the union bound the probability for one of these three terms to be congruent to some $x_p^{(j)} \pmod{p}$ is then smaller than $\frac{1}{2D}$. By another application of the union bound we see that there exists a choice of ε_i such that $f_{d-1}(xP_pQ - 1) \not\equiv 0 \pmod{p}$ holds for all non-exceptional $p \mid P$ and all $2D$ values of $x \in [1, Q)$ for which

$$x \equiv x_1^{(k)} + \sum_{i < |S'_2| \text{ odd}} \varepsilon_i \Delta_i \pmod{Q}$$

for some $1 \leq k \leq 2D$. Let us fix this choice of ε_i and denote the resulting values of x by $x^{(1)}, \dots, x^{(2D)}$.

As for the exceptional primes, we claim that for all distinct $k, k' \in \{1, \dots, 2D\}$ and all $p \mid P$ (so in particular for the exceptional primes) we have $x^{(k)} \not\equiv x^{(k')} \pmod{p}$. Indeed, $x^{(k)}$ and $x^{(k')}$ differ by tQ_{q_0} for some $0 < t < q_0$, while $p \nmid Q_{q_0}$ and

$$0 < t < q_0 \leq \frac{n}{2D} \leq \frac{n}{D+1} < p.$$

Since $f_{d-1} \pmod{p}$ has at most $D-1$ zeros for the at most two exceptional primes, while we have $2D$ values of $x^{(k)}$ to work with, we conclude that there must be at least two $x^{(k)}, x^{(k')}$ such that $f_{d-1}(xP_pQ - 1) \not\equiv 0 \pmod{p}$ for all $p \mid P$ and $x \in \{x^{(k)}, x^{(k')}\}$. This ensures the third property. As for the first property, $x^{(k)}$ and $x^{(k')}$ differ by at least Q_{q_0} , so at least one of them is larger than $Q_{q_0} > \frac{Q}{n}$. Hence, there exists an x satisfying all three properties, finishing the proof. $\square$

4. DECLARATION OF AI USAGE

Extending the author's construction of an x satisfying the first two properties of Theorem 3, ChatGPT 5.6-Sol Pro discovered how to apply [5, Theorem 1.4] to ensure that the third property holds as well. This paper is a human-written simplification of the proof that ChatGPT came up with. For the sake of transparency, the original AI-written output is still available at [2].

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GRONINGEN, THE NETHERLANDS

Email address: wonterm1@hotmail.com